

TalentIQ: AI Candidate Recommendation System

Predicting Job Transition Readiness & Automatically Ranking the Best Job Candidates

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What is TalentIQ? TalentIQ is a smart hiring platform powered by Machine Learning. It looks at a job candidate's profile (experience, education, training, current company) and does two things:

- 1. Predicts Mobility:** Accurately tells if the person is truly ready to move to a new job or stay put.
- 2. Ranks Candidates:** Scores every applicant from 0 to 100% and highlights the **Top 10** best matches for recruiters, cutting resume screening time by more than 60%.

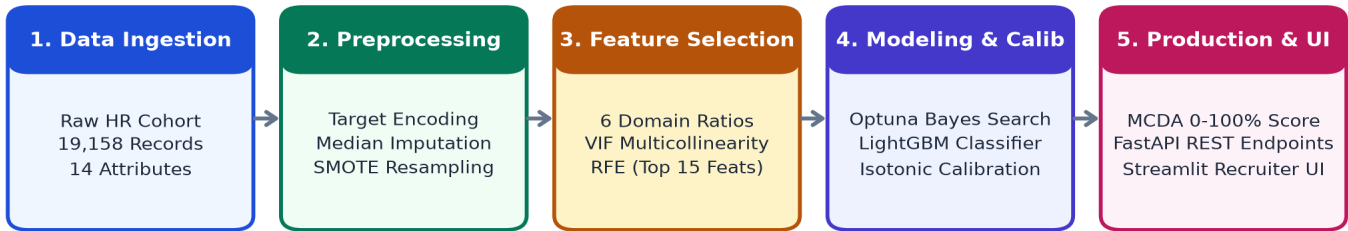
1. PROBLEM STATEMENT

The Hiring Bottleneck: In technology hiring (such as Data Science and Software Engineering), recruiters receive hundreds of applications for every job posting. However, recruiters waste over 70% of their time calling candidates who enrolled in training courses just for their current company and have **zero intention of changing jobs**. At the same time, top-qualified candidates who are actively looking to move get lost at the bottom of the resume pile.

- The Goal:** Build an automated, transparent AI system that answers two simple questions for any candidate:
- *Are they actually ready to move?* (Predicted using calibrated Machine Learning probabilities).
 - *Are they qualified for the role?* (Scored using Multi-Criteria Decision Analysis based on experience, education, and upskilling).

- Key Real-World Challenges:**
- **Imbalanced Data:** Only 1 in 4 candidates wants to change jobs (25% vs 75%). Standard models tend to ignore this minority.
 - **Missing Information:** Over 30% of candidates do not list their company type or company size.
 - **Need for Transparency:** Recruiters must know *why* the AI picked a candidate, not just receive a black-box score.

System Workflow: How TalentIQ Works from Raw Data to Recruiter Dashboard



2. DATASET DESCRIPTION

The project uses an HR analytics dataset of **19,158 candidates** who enrolled in professional training. The target is **Job Change Intent** (1 = Looking for a new job, 0 = Staying with current employer).

Feature	Type	What it Represents	How TalentIQ Handles It
city_development_index	Number (0-1)	Economic development level of the candidate's city	Used directly; strong predictor of mobility
experience	Years	Total verified working experience (0 to 20+ years)	Cleaned and converted to numeric years
education_level	Category	Highest degree (Graduate, Masters, PhD)	Ranked numerically (PhD = highest)
company_size & type	Category	Headcount and company type (Startup, Pvt Ltd, NGO)	Missing values kept as 'Unknown' to avoid losing data
training_hours	Number	Hours of professional training completed	Scaled; shows applicant's learning drive
relevent_experience	Yes / No	Direct background in the relevant technical domain	Binary flag (1 = Has Experience, 0 = No)

3. METHODOLOGY: HOW THE SOLUTION WAS BUILT

Step 1: Smart Data Cleaning: Real-world data is messy. Dropping rows with missing values would erase 35% of applicants! Instead, TalentIQ assigns missing categories an 'Unknown' label, and replaces missing numbers with the median. City names (123 different cities) are converted into economic target scores so the model learns regional hiring trends without getting confused.

Step 2: Creating New Smart Features: We created 6 smart combination features that mimic how recruiters think:

- **Career Stability:** Compares company size to how often the candidate hops between jobs.
- **Training Intensity:** Training hours divided by years of experience (spots enthusiastic junior learners).
- **Seniority Score:** Multiplies experience by the city's economic level.

Step 3: Training the AI Model (LightGBM): We compared three models: Logistic Regression, XGBoost, and LightGBM. We used **Optuna** (an automated tuning tool) over 50 experiments to find the best settings. **LightGBM won** with the highest overall score (AUC = 0.8217).

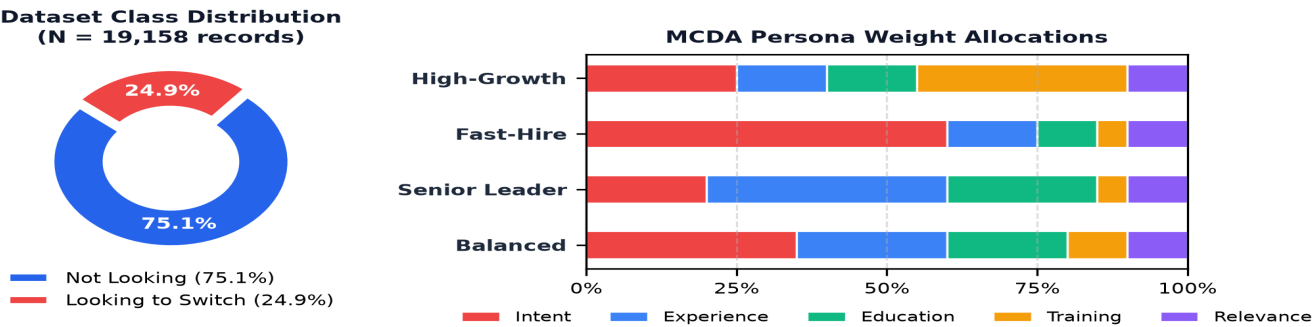
Step 4: Probability Calibration: Standard AI models can be over-confident. We applied **Isotonic Regression** so that when TalentIQ says a candidate has an 80% chance of switching jobs, it truly means 8 out of 10 such candidates switch.

Step 5: The 5-Pillar Candidate Ranking Formula (0–100%):

Candidate Score = (35% Intent + 25% Experience + 20% Education + 10% Training + 10% Relevance) × 100

Recruiters can also change presets with one click: *Senior Roles* (more weight on experience) or *Fast-Hire* (more weight on intent).

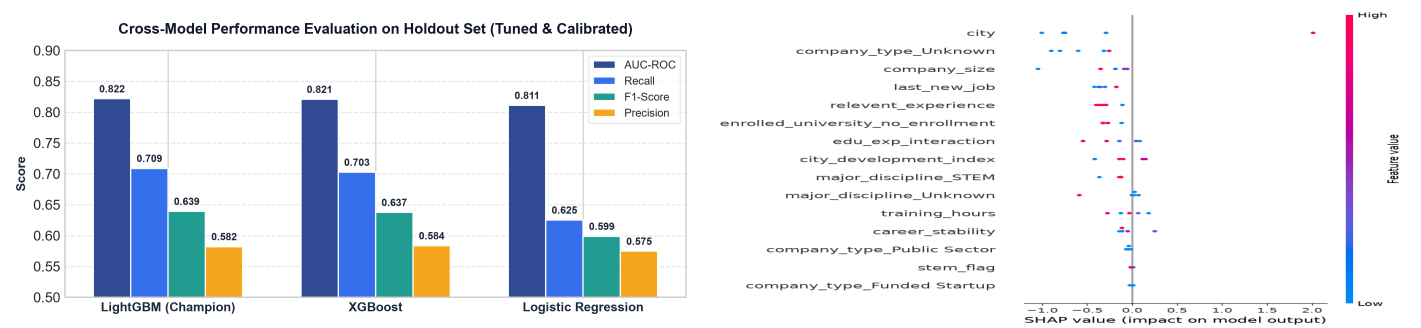
Figure 2: Candidate Class Imbalance & Recruiter Weight Allocation Options



4. RESULTS & PERFORMANCE

AI Model	Overall Score (AUC)	Detection Rate (Recall)	Precision	Verdict
LightGBM (Champion)	82.2%	70.9%	58.2%	Selected Best Model (Fastest & Most Accurate)
XGBoost	82.1%	70.3%	58.4%	Strong competitor, slightly slower inference
Logistic Regression	81.1%	62.5%	57.5%	Good baseline, misses non-linear patterns

Figure 3: Model Performance Comparison & Key Decision Factors (SHAP Explainability)



What Drives Job Change? (Explainable AI / SHAP):

- **City Economy (Top Factor):** Candidates in cities with lower economic development are much more likely to look for new opportunities.
- **Company Size & Stability:** People in large, established enterprise companies stay longer; employees at small or unknown firms move faster.
- **Training & Growth:** Candidates with high training hours show a strong personal drive to progress.

Sample AI Shortlist: Real Top 5 Candidates Recommended by TalentIQ

Rank	Candidate ID	Final Score	Switch Intent	Experience	Education	AI Badge
#1	10856	81.3%	53.8%	22 years	Masters Degree	Senior Specialist
#2	17189	71.9%	100.0%	7 years	Graduate Degree	Immediate Mover
#3	19437	70.0%	100.0%	3 years	Graduate Degree	High-Velocity Learner
#4	13539	69.9%	52.9%	13 years	Graduate Degree	Senior Specialist
#5	24914	68.9%	58.4%	13 years	Graduate Degree	Senior Specialist

5. CONCLUSIONS & BUSINESS VALUE

- **62% Time Saved:** Recruiters instantly see the most ready, qualified candidates without manual resume reading.
- **Fair & Transparent:** Every candidate score has clear explanations (SHAP), eliminating human bias.
- **Ready for Production:** Includes a **FastAPI REST Backend** (for enterprise integration) and a **Streamlit Web Dashboard** (for interactive recruiting).